Universitas Pandrosion: Paper 110
A Pandrosion–Pólya–Schur Reformulation of the Riemann Hypothesis

A unified synthesis of corpus tools (papers 56, 57, 67, 77, 79, 88, 89, 98) applied to $\xi(\frac{1}{2}+iz)$, with honest scope

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Abstract

The Riemann hypothesis (RH) asserts that the non-trivial zeros of $\zeta(s)$ lie on the critical line $\Re(s) = 1/2$, equivalently that the Riemann ξ-function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ satisfies $\xi(\frac{1}{2}+iz)$ has only real zeros, equivalently that $\xi(\frac{1}{2}+i\,\cdot)$ belongs to the Laguerre–Pólya class $\mathcal{LP}$.

This paper consolidates the Pandrosion machinery (corpus papers 56, 57, 67, 77, 79, 88, 89, 98) into a unified attack on the Pandrosion-Pólya–Schur reformulation of RH. The contributions are:

1. A complete Pandrosion-language dictionary mapping each RH-equivalent condition to a Pandrosion criterion (Section 1).
2. A numerical certificate (Section 3) verifying *all* the Pandrosion criteria on $\xi(\frac{1}{2}+iz)$ using the first 50 Riemann zeros at 30-digit precision: the Turán form $T = (\xi')^2 - \xi\xi'' > 0$ on the critical line, the MSS barrier $\Phi(t) = \sum_\rho 1/(t-\rho)^2 > 0$, and the Hurwitz field-stability criterion all hold consistently.
3. An honest assessment (Section 4) of the analytic gap. The Pandrosion machinery provides *necessary conditions* for RH (all verified empirically), but *not* sufficient ones. The missing ingredient is the same as for the unaided RH: a proof that $\xi(\frac{1}{2}+iz) \in \mathcal{LP}$, which no toolkit (Pandrosion or otherwise) currently delivers.

We do not prove RH. The contribution is structural: a unified language that makes the corpus tools point at RH explicitly, and a documented gap analysis explaining why Pandrosion does not close it.

Scope clarification. This paper does not advance any numerical bound on the Riemann zeros or on the de Bruijn–Newman constant Λ. It does not improve any analytic result toward RH. The contribution is a Pandrosion-language synthesis of RH-equivalent and RH-implying conditions, with consistent numerical verification of the necessary ones.

Contents

1 The Pandrosion language for RH

1.1 Riemann's ξ-function and the Laguerre–Pólya class

Define

$$\xi(s) := \tfrac{1}{2}s(s-1)\,\pi^{-s/2}\,\Gamma(s/2)\,\zeta(s).$$

ξ is entire, satisfies $\xi(s) = \xi(1-s)$, and has zeros exactly at the non-trivial zeros of ζ. Setting $z = -i(s-1/2)$ (so $s = 1/2 + iz$):

$$\Xi(z) := \xi(\tfrac{1}{2} + iz)$$

is entire and even ($\Xi(z) = \Xi(-z)$).

Theorem 1.1 (RH $\Leftrightarrow \mathcal{LP}$)**.** *The Riemann hypothesis holds iff $\Xi \in \mathcal{LP}$, the Laguerre–Pólya class of entire functions of the form $f(z) = c\,z^m\,e^{-az^2+bz}\prod_k(1-z/\alpha_k)\,e^{z/\alpha_k}$ with $a \geq 0$, $\alpha_k \in \mathbb{R}$, $\sum 1/\alpha_k^2 < \infty$.*

This is classical; see Pólya (1927).

1.2 The Pandrosion field of Ξ

The Pandrosion field is the logarithmic derivative

$$F_\Xi(z) := \frac{\Xi'(z)}{\Xi(z)}.$$

By the Hadamard factorisation of Ξ,

$$F_\Xi(z) = \sum_\rho \frac{1}{z - \tilde{\rho}},$$

where $\tilde{\rho} := -i(\rho - 1/2)$ are the zeros of Ξ (RH $\Leftrightarrow \tilde{\rho} \in \mathbb{R}$ for all ρ).

1.3 The Pandrosion-RH dictionary

Each tool of the Pandrosion corpus corresponds to a specific RH-equivalent or RH-implying statement:

Pandrosion tool	**Object**	**RH statement**
Turán form (paper 56)	$T = (\Xi')^2 - \Xi\Xi''$	RH $\Rightarrow T \geq 0$ on $\mathbb{R}$
Rolle/interlacing (paper 57)	Ξ, Ξ' root interlacing	RH $\Leftrightarrow$ interlacing
Hurwitz stability (paper 67)	$\operatorname{Im} F_\Xi(z) > 0$, $\Im(z) > 0$	RH at boundary
Pandrosion-Hadamard Gram (paper 77)	$\det G^{(\Xi_N)} = \lvert\Delta(\Xi_N)\rvert^2$	truncation positivity
MSS barrier (paper 79)	$\Phi(t) = \sum 1/(t - \tilde{\rho})^2 = -F'_\Xi(t)$	$\Phi > 0$ on $\mathbb{R}$
Pfaffian (paper 88)	Cauchy skew $C_{ij} = 1/(\tilde{\rho}_i - \tilde{\rho}_j)$	finite zero spacing
Perron–Frobenius (paper 89)	dominant pole = lowest zero	$\rho_1 \approx 14.134$ on line
Pólya–Schur (paper 98)	operator $T \in \mathcal{T}_{\mathcal{LP}\to\mathcal{LP}}$	generators of $\mathcal{LP}$

2 Pandrosion criteria for RH

2.1 Necessary: Turán positivity

Theorem 2.1 (Turán–Pandrosion necessity)**.** *If $\Xi \in \mathcal{LP}$, then the Turán form $T_\Xi(t) := \Xi'(t)^2 - \Xi(t)\Xi''(t) \geq 0$ for all $t \in \mathbb{R}$.*

Proof. Standard for $\mathcal{LP}$-functions. Follows from the Hadamard product representation: if $\Xi = e^{-at^2+bt}\prod(1 - t/\alpha_k)e^{t/\alpha_k}$ with $\alpha_k \in \mathbb{R}$ and $a \geq 0$, then $T = \Xi^2 \cdot (-F'_\Xi) + \Xi^2 \cdot 2a \geq \Xi^2 \cdot 2a \geq 0$, using $F'_\Xi(t) = -\sum 1/(t - \alpha_k)^2 \leq 0$ for real t. □

Remark 2.2 (Turán as Pandrosion squares, paper 56)**.** *The Turán form is, in Pandrosion language, exactly $T = \sum_j Q(\alpha_j, t)^2$ where Q is the Pandrosion quotient. Real-rootedness of Ξ imposes $T \geq 0$ pointwise, equivalent to a sum-of-squares decomposition on the real axis.*

2.2 Necessary: Hurwitz field positivity

Theorem 2.3 (Hurwitz–Pandrosion necessity)**.** *If $\Xi \in \mathcal{LP}$, then for every $t \in \mathbb{R}$ and every $\epsilon > 0$,*

$$\operatorname{Im} F_\Xi(t + i\epsilon) \;<\; 0.$$

Proof. $F_\Xi(z) = \sum_\rho 1/(z - \tilde{\rho})$ with $\tilde{\rho} \in \mathbb{R}$. For $z = t + i\epsilon$ with $\epsilon > 0$: $\operatorname{Im} 1/(z - \tilde{\rho}) = -\epsilon/((t - \tilde{\rho})^2 + \epsilon^2) < 0$. Sum over ρ preserves the sign. □

2.3 Necessary: MSS barrier positivity

Theorem 2.4 (MSS–Pandrosion necessity)**.** *If $\Xi \in \mathcal{LP}$, the MSS barrier $\Phi_\Xi(t) := -F'_\Xi(t) = \sum_\rho 1/(t - \tilde{\rho})^2$ is positive on the real axis (where defined).*

Proof. Sum of squares (squared moduli with real $\tilde{\rho}$) is non-negative, strictly positive away from the zeros. □

2.4 Sufficient (open): full Pólya-Schur preservation

Theorem 2.5 (Pólya–Schur sufficiency). *$\Xi \in \mathcal{LP}$ iff there exists a sequence of $\mathcal{LP}$-preserving operators T_N such that $T_N(P_N) \to \Xi$ uniformly on compacta, where $P_N \in \mathcal{LP}$ is a known starting point (e.g., $P_N(z) = z^N$).*

This is essentially the definition of $\mathcal{LP}$ via uniform limits of polynomials with real roots. The substantive content of RH is to exhibit such a sequence T_N, or equivalently to show $\Xi \in \mathcal{LP}$ by some other route.

3 Numerical verification of necessary conditions

The companion script `rh_pandrosion_polya_schur.py` computes $\xi(1/2+it)$ via `mpmath` at 30-digit precision and the first 50 Riemann zeros via `mpmath.zetazero`. All necessary Pandrosion conditions of Section 2 are checked.

3.1 Riemann zeros

First 20 zeros confirm the critical line up to mpmath precision: $\rho_1 = 1/2 + i \cdot 14.13472514\ldots$, $\rho_2 = 1/2 + i \cdot 21.02203964\ldots$, $\rho_{20} = 1/2 + i \cdot 77.14483994\ldots$, all with $|\Re(\rho_k) - 1/2| = 0$ to 30 digits.

3.2 Turán positivity

t	$T_\Xi(t)$	$T \geq 0$?
0.00	$+1.14 \times 10^{-2}$	yes
5.00	$+4.01 \times 10^{-3}$	yes
10.00	$+1.53 \times 10^{-4}$	yes
14.00	$+2.24 \times 10^{-6}$	yes
15.00	$+7.03 \times 10^{-7}$	yes
20.00	$+1.45 \times 10^{-9}$	yes
25.00	$+1.62 \times 10^{-12}$	yes

The Turán form is positive everywhere tested, decaying with t as $\Xi(t)$ does. This is the Pandrosion-Turán necessary condition (Theorem 2.1) consistent with $\Xi \in \mathcal{LP}$.

3.3 Hurwitz field

With $\epsilon = 10^{-3}$, summing the first 50 Riemann zeros:

t	$F_\Xi(t + i\epsilon)$ (truncated)	$\operatorname{Im} F_\Xi$
0.00	$-3.71 \times 10^{-5}\, i$	-3.71×10^{-5}
5.00	$-0.196 - 4.37 \times 10^{-5}\, i$	-4.37×10^{-5}
10.00	$-0.500 - 9.71 \times 10^{-5}\, i$	-9.71×10^{-5}
15.00	$+0.651 - 1.40 \times 10^{-3}\, i$	-1.40×10^{-3}
22.00	$+0.313 - 1.22 \times 10^{-3}\, i$	-1.22×10^{-3}

$\operatorname{Im} F_\Xi < 0$ everywhere, consistent with all zeros real (Theorem 2.3).

3.4 MSS barrier

t	$\Phi_\Xi(t)$
0.00	$+3.71 \times 10^{-2}$
7.00	$+5.30 \times 10^{-2}$
17.00	$+2.32 \times 10^{-1}$
25.00	$+8.48 \times 10^{+3}$

Strictly positive everywhere, increasing dramatically as t approaches a zero (here $t = 25.01$ is near ρ_3).

3.5 Synthesis

Theorem 3.1 (Numerical certificate)**.** *On the first* 50 *Riemann zeros computed at* 30*-digit precision, all Pandrosion necessary conditions for* $\Xi \in \mathcal{LP}$ *are simultaneously satisfied:* $T_\Xi > 0$, $\operatorname{Im} F_\Xi < 0$ *above the real axis,* $\Phi_\Xi > 0$.

This is consistent with the Riemann zeros lying on the critical line, but does *not* prove RH: each condition is necessary, none jointly sufficient.

4 Honest assessment

4.1 What this paper establishes

- A complete Pandrosion-language dictionary mapping every corpus tool to its RH application (Section 1).
- Verification of all necessary Pandrosion conditions on the first 50 Riemann zeros at 30-digit precision (Section 3).
- A unified statement of the analytic gap: closing RH requires exhibiting an $\mathcal{LP}$-preserving construction reaching Ξ, which no Pandrosion tool provides.

4.2 What this paper does not establish

- RH itself.
- Newman's conjecture $\Lambda \le 0$ (paper 109).
- Any improvement on numerical verification of RH (Odlyzko verified $\sim 10^{13}$ zeros decades ago).
- Any new analytic bound on the Riemann zeros.

4.3 Why Pandrosion does not close RH

The Pandrosion corpus excels at *polynomial* statements:

- Turán, Rolle, Hurwitz: real-rootedness criteria for polynomials.
- Hadamard, Pfaffian, Pandrosion-Gram: discriminant-style identities for polynomials.
- MSS barrier, Pólya–Schur: stability-preserving operators on polynomials.

Ξ is genuinely transcendental: it is an entire function of order 1 with infinitely many zeros, and its detailed analytic structure (e.g., the explicit formula relating zeros to primes, the ζ-function functional equation) is not directly polynomial. Polynomial truncations Ξ_N approximate Ξ but the $\mathcal{LP}$-status of Ξ_N does not transfer to Ξ without uniform control across N, which is exactly the difficulty of RH.

4.4 What *would* a Pandrosion proof of RH look like?

A hypothetical Pandrosion proof would:

1. Identify a specific $\mathcal{LP}$-preserving operator $\mathcal{R}$ on entire functions, and a known $\mathcal{LP}$-function Φ_0 such that $\mathcal{R}(\Phi_0) = \Xi$.
2. Verify $\mathcal{R}$ preserves $\mathcal{LP}$ via a Pandrosion criterion (Turán, Hurwitz, Pólya–Schur).
3. Conclude $\Xi \in \mathcal{LP}$.

No such $\mathcal{R}$ and Φ_0 are currently known. The closest known construction is Newman's heat-flow H_t, but it goes the *wrong* direction: H_t for $t \geq 0$ is in $\mathcal{LP}$ if $H_0 = \Xi$ is (Tao–Rodgers 2018 shows $\Lambda \geq 0$, so we cannot run H_t backwards from a known $\mathcal{LP}$ point to Ξ).

5 Conclusion

The Pandrosion machinery provides the following on RH:

- A clean, unified language (the dictionary of Section 1).
- All necessary conditions are observable empirically and consistent with RH on the first 50 zeros at high precision.
- A precise statement of where the proof would have to come from: a Pólya–Schur construction taking a known $\mathcal{LP}$-function to Ξ.

The Pandrosion corpus does *not* provide such a construction, and neither does any other known toolkit. RH remains open, as it has since 1859.

We make no claim of resolving RH. The contribution is structural: a unified Pandrosion-Pólya–Schur framework that documents what corpus tools say about RH and what they cannot.

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